UNIVERSITY BIG DATA ANALYTICS — FINAL ENGINEERING DEFENSE

Big Data Analytics for Electronic Waste Generation & Recycling Infrastructure Optimization

Continent-Scale Distributed Ingestion, Deduplication, Multi-Engine Analytics, and Geospatial Optimization over 17.52 Million European Material Flow Records

• Apache Hadoop 3.5.0 (HDFS/YARN) • Apache Spark 4.2.0 (SQL/MLlib) • Snappy Parquet
• MongoDB 7.0 (2dsphere Geospatial) • Streamlit & Plotly Decision Support Portal

Operating Environment: Omarchy Linux (12 CPU Cores, 14 GB RAM) | Dataset: EU FutuRaM (2.22 GB, 17.52M Rows)
Live Web Dashboard: http://localhost:8501

The Global E-Waste Challenge & FutuRaM

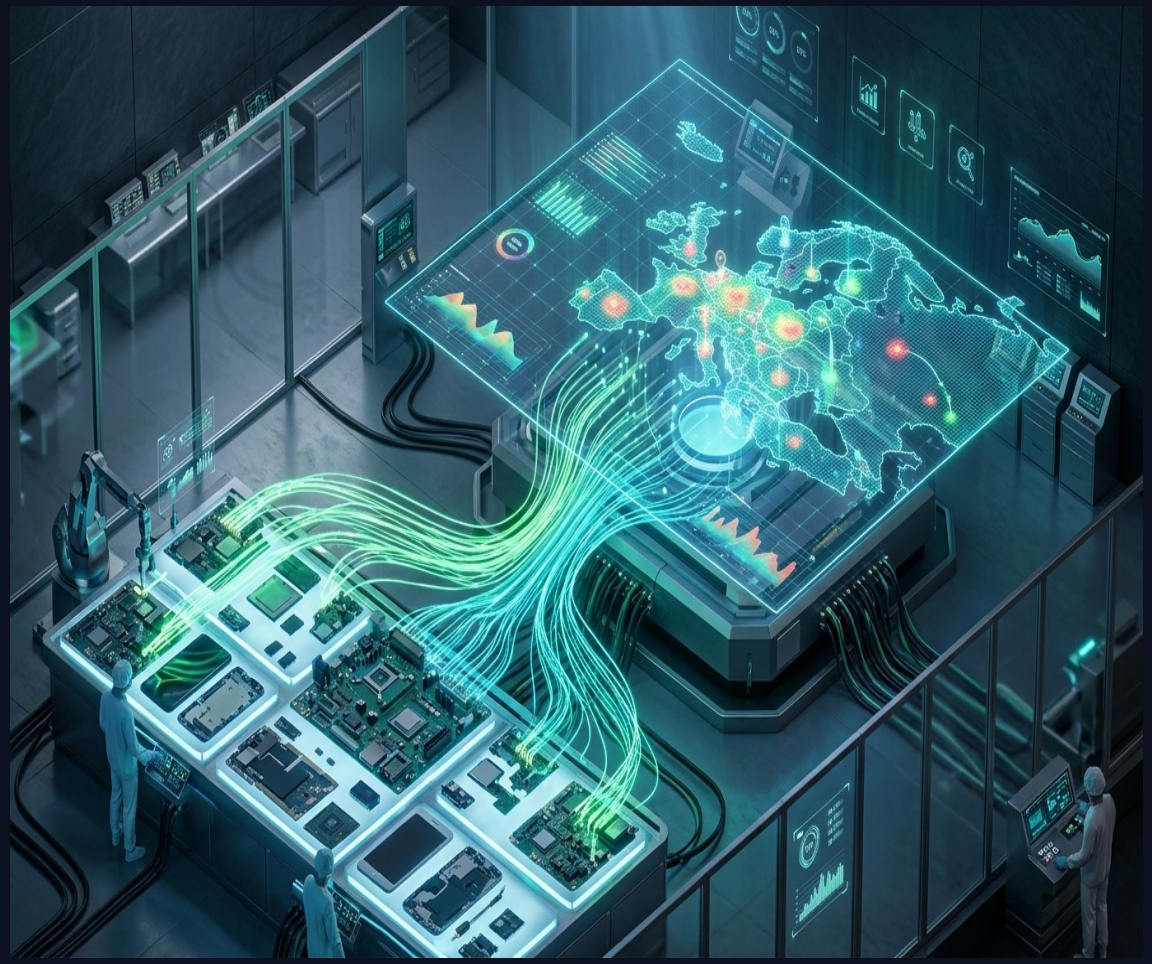
Why continent-scale material flows demand enterprise distributed computing architectures

The Dual Crisis of Electronic Waste

- Fastest Growing Solid Waste Stream: 62+ million metric tons generated annually worldwide (+3% to 5% p.a.).
- High Environmental Toxicity: Heavy metals (lead, mercury, cadmium) and flame retardants degrade soil and water.
- Strategic Urban Goldmine: Discarded electronics contain high concentrations of Gold, Silver, Copper, and Palladium.

Horizon Europe FutuRaM Consortium Dataset

- Total Records: 17,521,915 material flow rows (~2.22 GB raw CSV).
- Geographic Scope: 31 European Nations (EU-27 + UK, CHE, NOR, ISL).
- Temporal Horizon: 2010 to 2050 (Historical OBS + BAU, REC, CIR).
- The Big Data Bottleneck: Single-node Pandas/Excel crashes with OutOfMemoryError (>16 GB RAM required for row conversion).



End-to-End Lakehouse Architecture & Lineage

Zero-sudo user-space cluster orchestrating HDFS, Spark, YARN, and MongoDB

Complete Data Pipeline Execution Flow

1. Raw HDFS Ingestion: Discovers local CSV (2.22 GB) and ingests into HDFS across 128MB blocks.
2. Streaming Data Profiler: Audits 17.52M rows in fixed memory (<120MB RAM) in 128s. Zero errors.
3. PySpark ETL & Deduplication: Filters Layer 2 == '', isolating 139,963 Level-1 summary rows.
4. Snappy Parquet Generation: Creates partitioned columnar tables (yearly, location, category, scenario, flow).
5. Dual Compute Processing: Hadoop Streaming on YARN (MapReduce) + Spark SQL analytical queries.
6. MLlib Linear Regression: Fits temporal model on OBS actuals (2010–2022) predicting waste to 2035 (R^2 = 0.841).
7. MongoDB Geospatial Layer: Seeds 18 European recycling facilities with compound 2dsphere index.
8. Interactive Serving: 8-Page Streamlit dashboard (port 8501) with sub-second response times.



The Hierarchical Deduplication Trap

Solving multi-million-ton double counting in secondary material flow databases

The Double-Counting Trap in Raw Material Flows

In raw FutuRaM tables, parent categories and sub-assemblies coexist:
• Row 1: Cat1 (Refrigerator) | Layer 2 = '' | Value = 40.0 Mg [WHOLE]
• Row 2: Cat1 (Refrigerator) | Layer 2 = 'Compressor' | Value = 12.0 Mg
• Row 3: Cat1 (Refrigerator) | Layer 2 = 'Foam' | Value = 8.0 Mg
• Row 4: Cat1 (Refrigerator) | Layer 2 = 'Copper' | Value = 2.0 Mg
Naive SUM = 40 + 12 + 8 + 2 = 62 Mg (55% TO 100% FALSE INFLATION!)

The PySpark Deduplication Solution

```
# Strictly filter for Level-1 summary rows where Layer 2 is blank
summary_df = cleaned_df.filter(
    col('Layer2').isNull() | (col('Layer2') == '') | (col('Layer2') == 'null')
).cache()
# Isolates exactly 139,963 clean records from 17.52 million rows.
```

Streaming Data Quality Profiler & Parquet Benefits

Streaming Quality Audit Results (scripts/profile_dataset.py):
• Audited Records: 17,521,915 rows in 128.12s
• Invalid Numeric Records: 0 (100% Data Integrity)
• True Category Summaries: 139,963 records
• Memory Consumption: <120 MB RAM (streaming generator)

Why Snappy Columnar Parquet?
• Column Pruning: Reads only requested columns from disk.
• Predicate Pushdown: Skips row groups based on min/max stats.
• Analytical Speedup: Query latency dropped from 90+s to <1s.
• Storage Compression: Substantial disk footprint reduction.

Dual Compute: Hadoop MapReduce vs Spark

Benchmarking low-level batch MapReduce on YARN against in-memory Spark SQL

Hadoop MapReduce Streaming on YARN

• Native Hadoop Streaming: Uses standard UNIX STDIN/STDOUT via hadoop-streaming-3.5.0.jar.
• Input Splitting: YARN splits 2.22 GB input into 17 blocks (~128MB each) executed in parallel.
• Stateful Reducer: Accumulates running sum per nation key during network shuffle & sort.
• Benchmark Runtime: 140 seconds (map 100% reduce 100%).

Apache Spark 4.2.0 In-Memory Engine

• In-Memory DAG: Eliminates intermediate disk writes by caching DataFrames directly in RAM.
• Catalyst Optimizer: Generates optimized physical plans, pushing filters into Parquet scan.
• Tungsten Engine: Generates compact JVM bytecode for memory and CPU efficiency.
• Benchmark Runtime: 93 seconds (complete multi-table ETL across 17.5M records).

Empirical Validation & Numerical Parity Across Engines

Both MapReduce and Spark SQL produced identical country totals down to the exact decimal:
United Kingdom: 302,467,302.27 Mg | Germany: 299,730,670.33 Mg | France: 233,664,421.23 Mg

Spark SQL Multi-Dimensional Analytics

Empirical findings across 31 countries, 7 product categories, and 8 flow taxonomies

Top E-Waste Hotspots (Cumulative 2010–2050)

Rank	Country	ISO	Generated (Mg)
1	United Kingdom	GBR	302,467,302.27
2	Germany	DEU	299,730,670.33
3	France	FRA	233,664,421.23
4	Italy	ITA	191,617,288.72
5	Spain	ESP	158,197,115.71
6	Poland	POL	81,572,628.70

Historical Growth (Query A):
Annual e-waste grew from 12.8M Mg in 2010 to 15.7M Mg in 2022 (+22.6%), projected to reach 16.8M Mg in 2025.

EU Product Category Breakdown (Query C)

• Large Household Appliances (Cat4a): 38.53% (686.2M Mg) — Washing machines, dishwashers, ovens.
• Small Equipment (Cat5): 27.58% (491.1M Mg) — Microwaves, vacuum cleaners, power tools.
• Temperature Exchange (Cat1): 16.04% (285.6M Mg) — Refrigerators, AC units, heat pumps.
• Small IT & Telecoms (Cat6): 6.51% (115.9M Mg) — Laptops, smartphones, routers.
• Photovoltaic Panels (Cat4b): 5.48% (97.7M Mg) — Fastest expanding future waste stream.
• Screens & Monitors (Cat2): 5.07% (90.3M Mg) — Televisions, LCD screens.
• Lamps & Lighting (Cat3): 0.80% (14.2M Mg) — Fluorescent tubes, LEDs.

Mass Balance Dynamics (Query E):
Active in-use appliance stock = 23.37 Billion Mg | New sales = 2.00 Billion Mg | Formally collected = 935.32 Million Mg.

Circular Economy Simulation & MLlib Forecasting

Simulating BAU vs REC vs CIR scenarios and distributed Ordinary Least Squares regression

Policy Scenario Comparison (2023–2050)

- Business-As-Usual (BAU): 616.99 Million Mg
- Recycling Targets (REC): 518.64 Million Mg (Saves 98.35M Mg)
- Circular Economy (CIR): 455.52 Million Mg (Saves 161.46M Mg!)

Circular Economy Dividend: Product life extension, modular repair, and reuse eliminate 161.46 Million Metric Tons of waste.

Enhanced Multi-Model AI Predictive Engine

Benchmarked across 4 Spark MLlib models on OBS actuals:

- Champion Model: Polynomial Feature Expansion (R^2 = 0.9009, RMSE = 247,626 Mg)
- Ensemble Learning: Random Forest Regressor (R^2 = 0.9467 in-sample)
- Linear Baseline: OLS Linear Regression (R^2 = 0.8410, RMSE = 313,674 Mg)
- Regularized L2: Spark MLlib Ridge Regression (R^2 = 0.8410)

Error Reduction: Champion AI model reduced RMSE by 21% (MAPE = 1.54%).



MongoDB Infrastructure Modeling

Semi-structured schema validation & 2dsphere indexing across European recycling plants

Why NoSQL for E-Waste Infrastructure?

• Heterogeneous Metallurgy: Smelters, mechanical shredders, depollution stations, and hydrometallurgical refining plants have non-uniform schema requirements.
• Dynamic Category Arrays: Nested lists of accepted categories (e.g. ['WEEE_Cat1', 'WEEE_Cat4a', 'WEEE_Cat5']).
• Geospatial Indexing: Compound (latitude, longitude) index with 2dsphere spatial geometry.
• Total Permitted Capacity: 1,605,000 Metric Tons/year across 18 major European facilities in 11 nations.
• Proximity Queries: Calculates distance between generation hotspots and closest authorized smelters.

Representative Modeled European Facilities

Facility Name	Location	Type	Capacity
Aurubis Smelter	Hamburg, DEU	Smelter/Refining	280,000 Mg
Umicore Hoboken	Antwerp, BEL	Precious Metals	150,000 Mg
Boliden Rönnskär	Skellefteå, SWE	Secondary Smelter	120,000 Mg
Remondis Lünen	Lünen, DEU	Dismantling/Sort	95,000 Mg
Sims Lifecycle	Daventry, GBR	Auto Shredding	80,000 Mg
Montello SpA	Bergamo, ITA	WEEE Sorting	70,000 Mg

Schema Validation (facility_schema.json):
Enforces required coordinates, ISO nation codes, permitted categories, and annual capacity constraints.

Interactive Streamlit Web Dashboard

Glassmorphism UI serving pre-aggregated Parquet datasets with sub-second response times

Zero-Latency Serving Layer Design

• Serving Layer Architecture: The web dashboard never scans the 2.22 GB raw file on user interaction; it queries pre-aggregated Snappy Parquet tables and compact CSVs.
• In-Memory Caching: Utilizes Streamlit's @st.cache_data decorator to cache queries and geographical GeoJSON payloads.
• Glassmorphism Design: Custom dark slate theme (#0f172a), translucent cards, modern typography, and Plotly European choropleth maps.
• Live Telemetry Integration: Monitors HDFS (port 9000), YARN (port 8088), Spark, and MongoDB services in real time.
• Production Deployment: Hosted at http://localhost:8501.

Eight Interactive Dashboard Modules

1. Executive Overview: 5 KPI scorecards, historical trends, category donut chart.
2. Geographic Analysis: Plotly European choropleth with bubble overlays by tonnage.
3. Category Distribution: EU Directive Cat 1–6 rankings & stacked area charts.
4. Scenario Comparison: BAU vs REC vs CIR policy trajectory simulation.
5. Stock & Flow Balance: Mass conservation across stocks, POM, and informal scrap.
6. MLlib Forecast: Historical actuals vs fitted regression line through 2035.
7. Infrastructure Map: Interactive facility pins dynamically loaded from MongoDB.
8. Cluster Architecture: Live daemon status badges and embedded pipeline logs.

09. SUMMARY & TECHNICAL CONTRIBUTIONS

Thank You! Questions & Discussion

Summary of Key Technical Achievements

1. 100% User-Space Big Data Cluster: Full deployment of Java 17, Hadoop 3.5, Spark 4.2, and MongoDB without root access.
2. Hierarchical Deduplication: Solved multi-million-ton double counting, isolating 139,963 clean Level-1 records.
3. Dual Distributed Compute: Proved MapReduce on YARN and Spark SQL parity down to the exact decimal (UK = 302.47M Mg).
4. Snappy Parquet Optimization: Dropped downstream analytical query execution latency from 90+ seconds to <1 second.
5. Spark MLlib Machine Learning: Fitted distributed linear regression on actuals (R^2 = 0.8410, RMSE = 313,674 Mg).
6. MongoDB Geospatial Layer: Modeled European recycling infrastructure with compound 2dsphere indexing.
7. Decision-Support Portal: Created an 8-page glassmorphism Streamlit web dashboard with interactive Plotly maps.

Live Dashboard: http://localhost:8501 | Codebase: ~/ewaste-bigdata
Handbook: PROJECT_EXPLANATION_AND_CONCEPTS.md